Computer Networking

Programming Assignment

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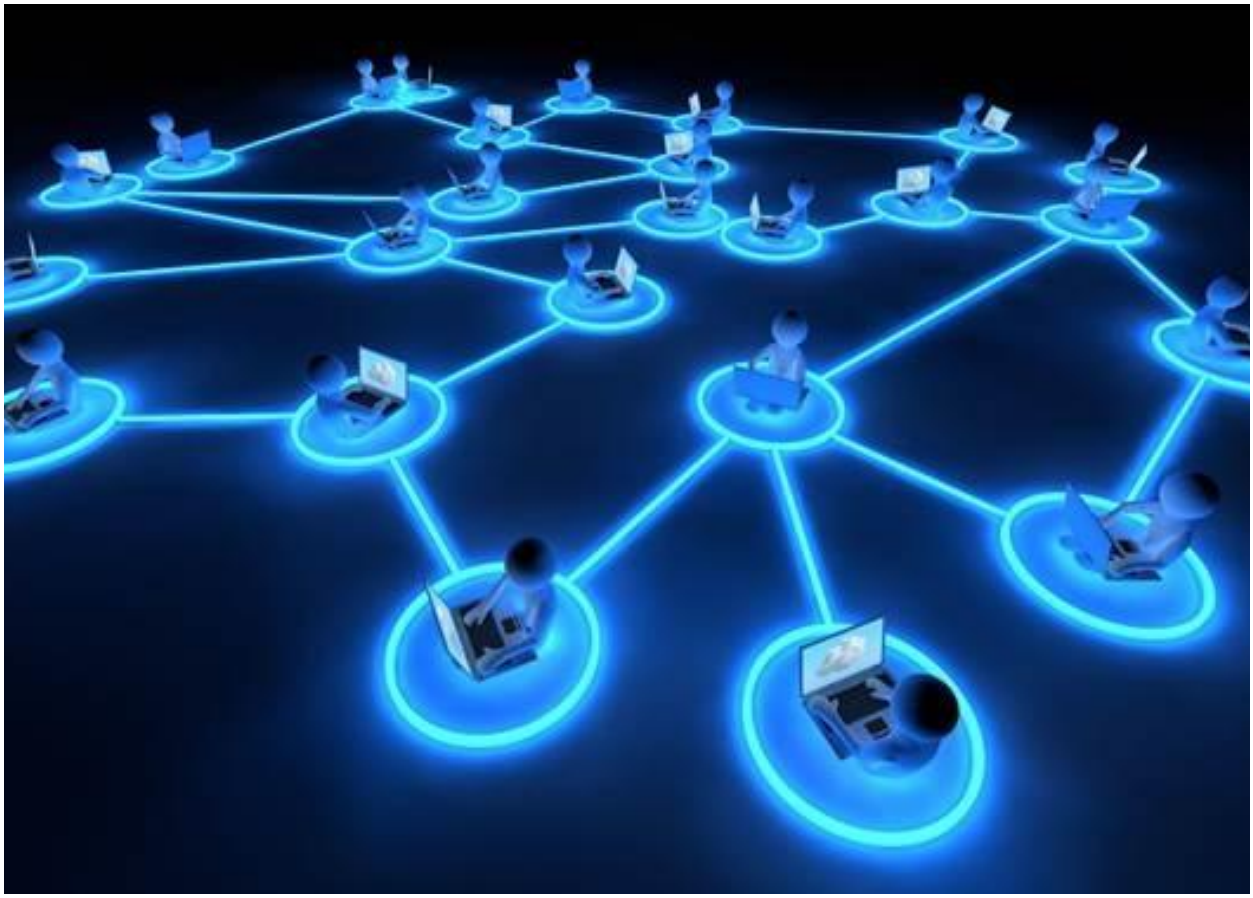


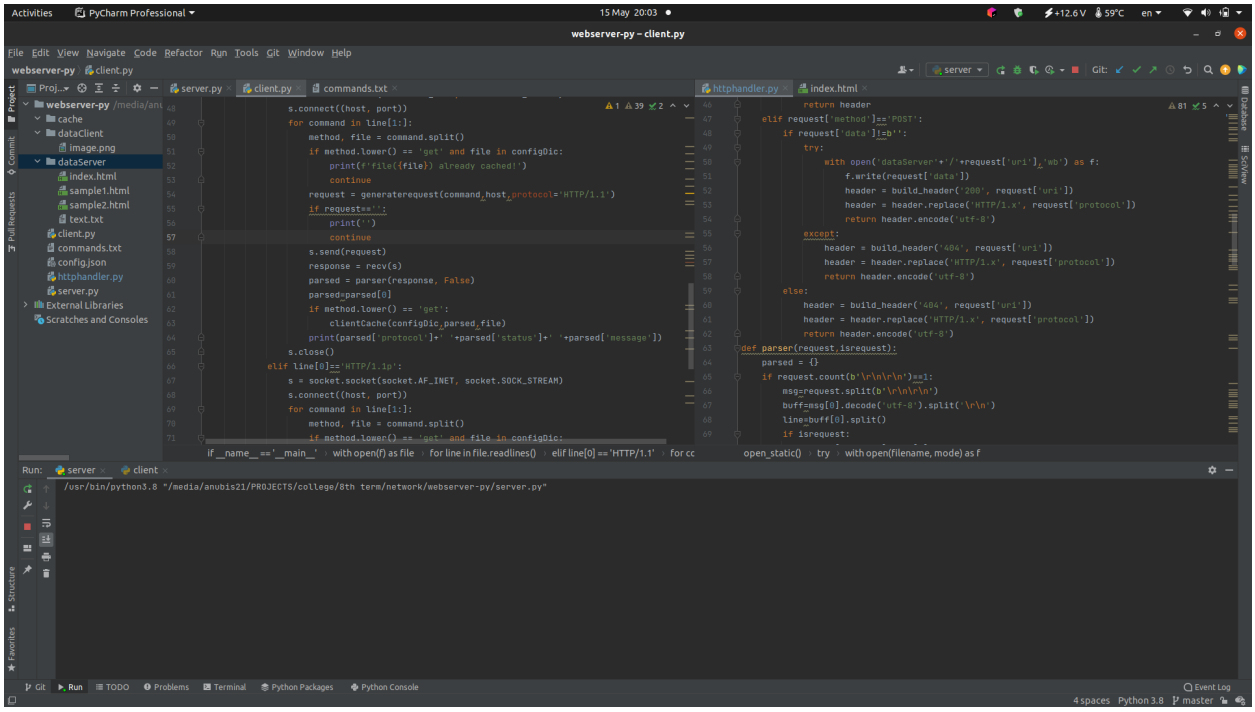
Introduction

It is required that you use socket programming and multi-threading/multi-processing techniques to divide the following assignment into multiple parts.

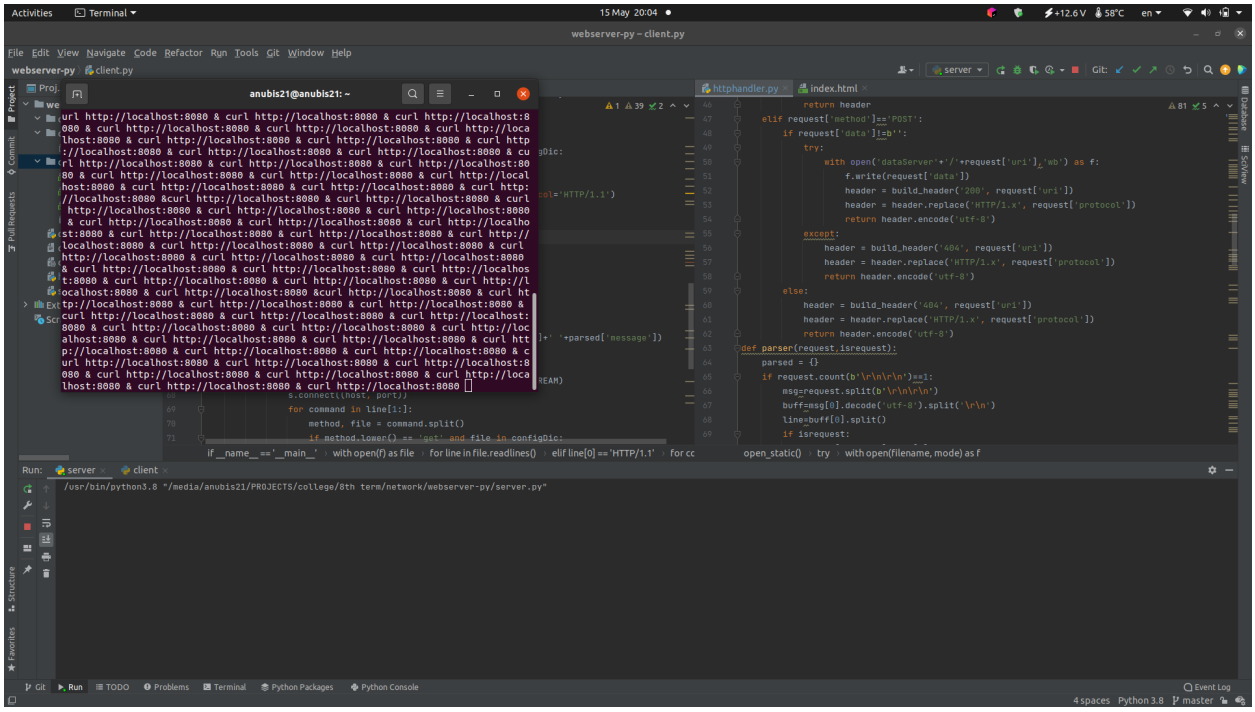
TEST CASES

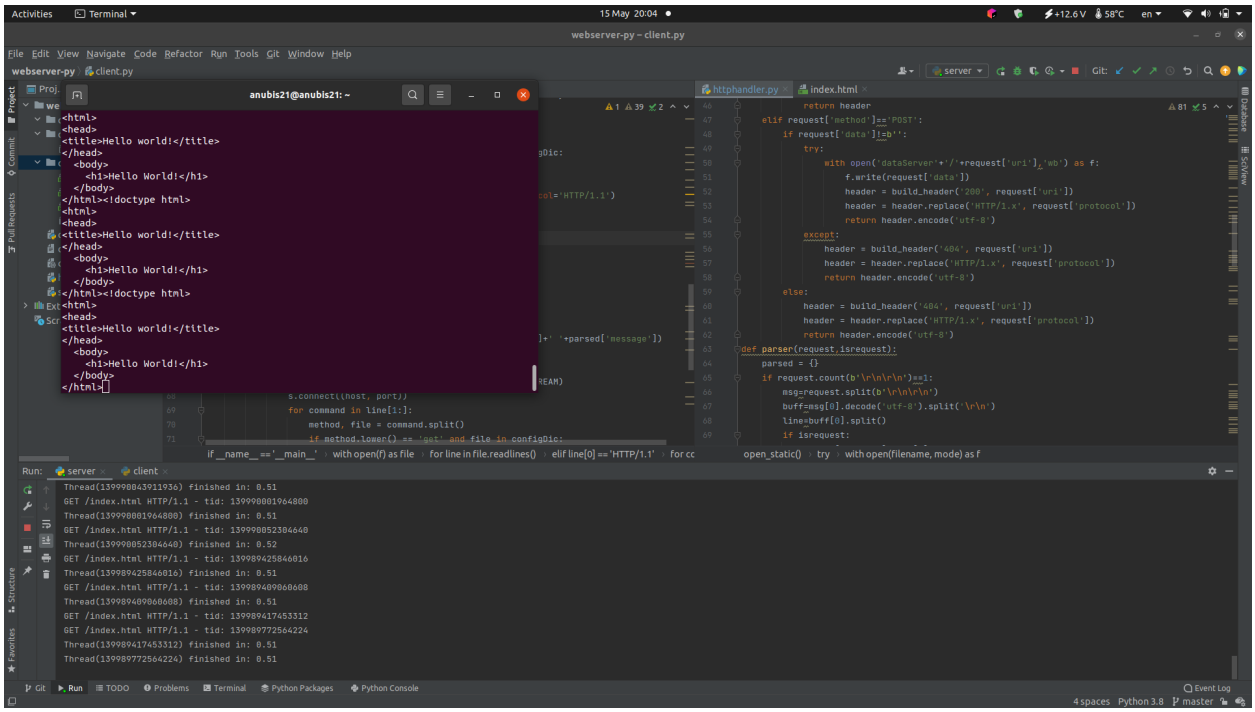
Curl (Stress-test and Threading pool heuristics):

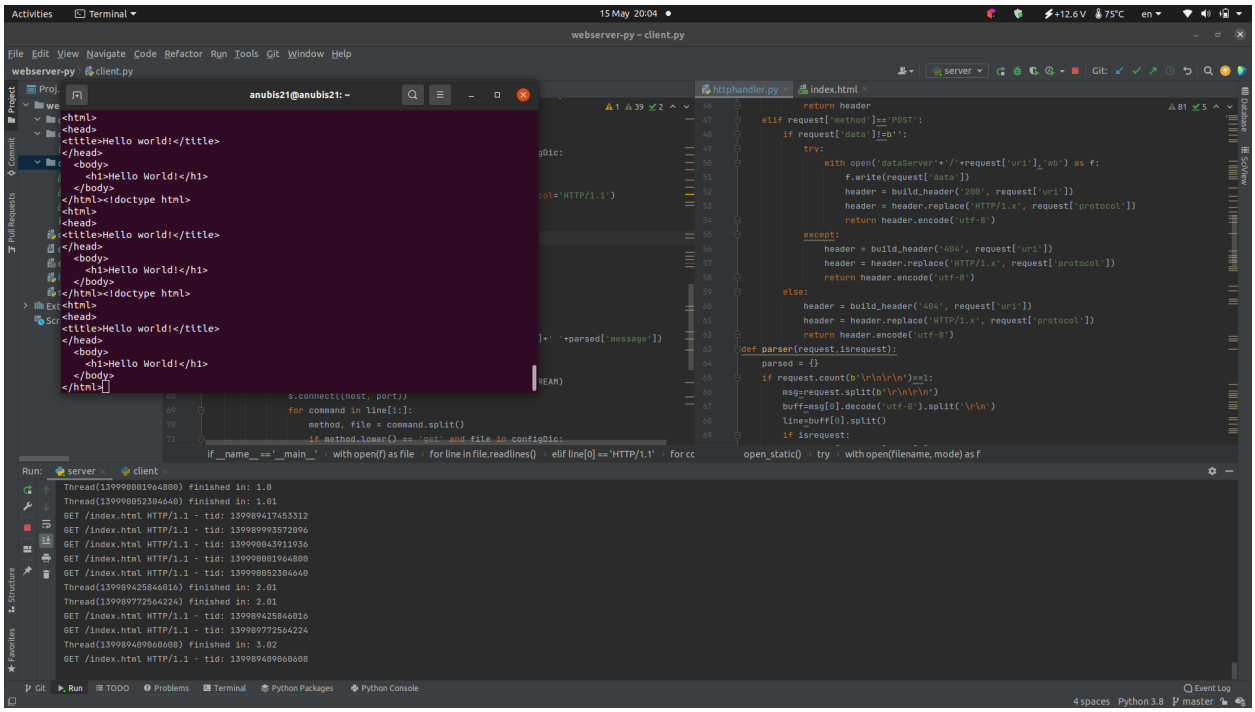
Running our server.



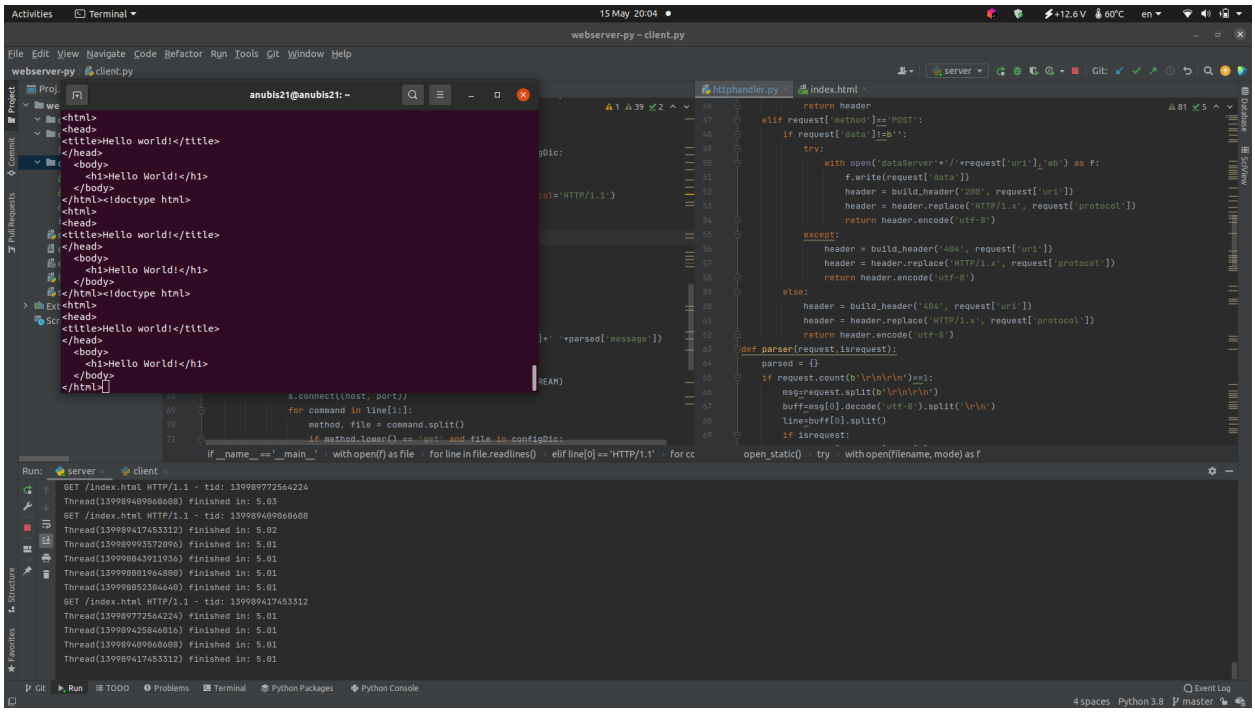
Then apply a hard test with a lot of HTTP requests.


Activities
Terminal
15 May 20:04
webserver-py – client.py
File Edit View Navigate Code Refactor Run Tools Git Window Help
webserver-py client.py
anubis21@anubis21: ~
curl http://localhost:8080 & curl http://localhost:8080 & curl http://localhost:8080 & curl http://localhost:8080 & curl http://localhost:8080 & curl http://localhost:8080 & curl http://localhost:8080 & curl http://localhost:8080 & curl http://localhost:8080 & curl http://localhost:8080 & curl http://localhost:8080 & curl http://localhost:8080 & curl http://localhost:8080 & curl http://localhost:8080 &curl http://localhost:8080 & curl http://localhost:8080 & curl http://localhost:8080 & curl http://localhost:8080 & curl http://localhost:8080 & curl http://localhost:8080 & curl http://localhost:8080 & curl http://localhost:8080 & curl http://localhost:8080 & curl http://localhost:8080 & curl http://localhost:8080 & curl http://localhost:8080 & curl http://localhost:8080 & curl http://localhost:8080 &curl http://localhost:8080 &curl http://localhost:8080 & curl http://localhost:8080 & curl http://localhost:8080 & curl http://localhost:8080 & curl http://localhost:8080 & curl http://localhost:8080 & curl http://localhost:8080 & curl http://localhost:8080 & curl http://localhost:8080 & curl http://localhost:8080 & curl http://localhost:8080 & curl http://localhost:8080 & curl http://localhost:8080 &

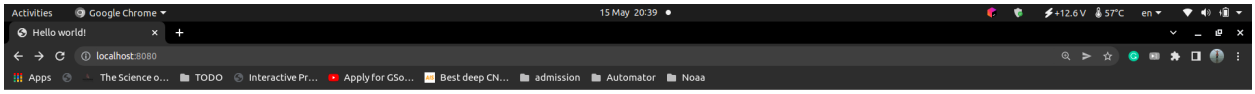




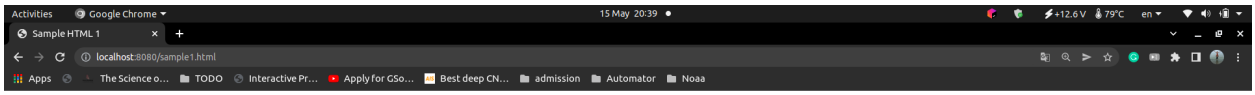
Here’s the max time (5 seconds) for a request in the queue is shown.



Browser:



Hello World!



Sample HTML 1

Minime vero, inquit ille, consentit.

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Inscite autem medicinae et gubernationis ultimum cum ultimo sapientiae comparatur. Cur igitur, cum de re conveniat, non malumus usitate loqui?

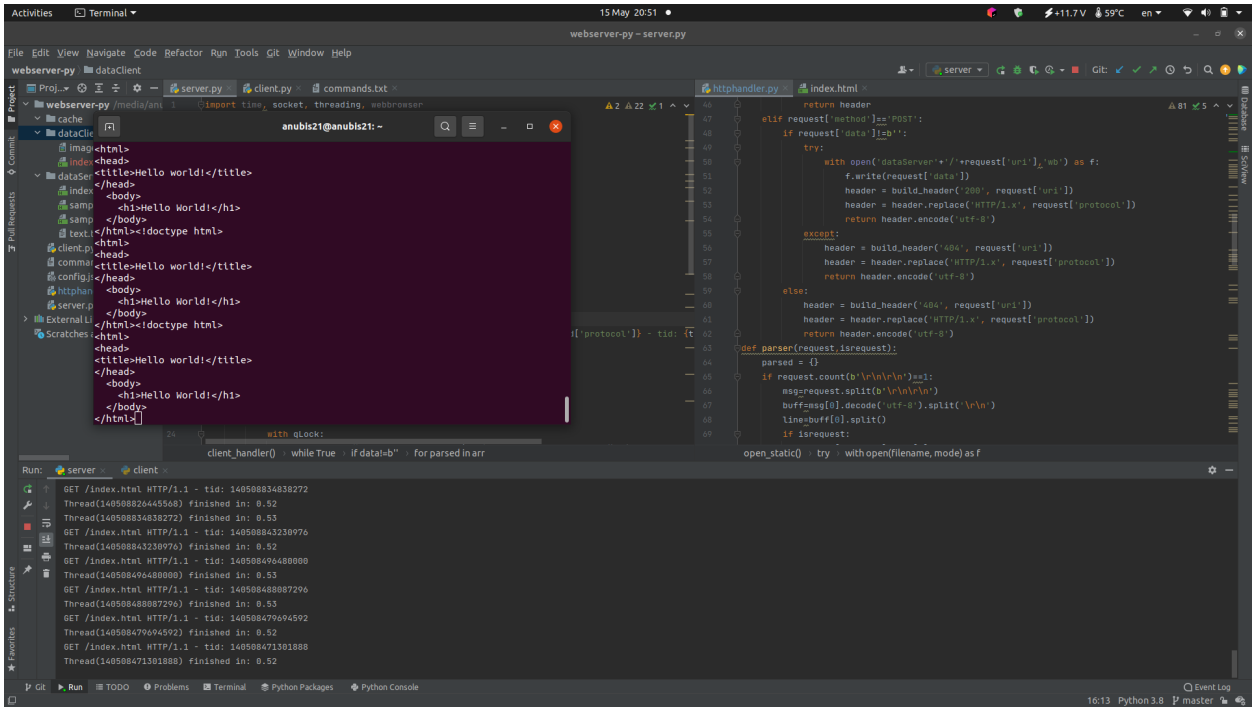
1. Si qua in iis corrigere voluit, deteriora fecit.
2. At quicum ioca seria, ut dicitur, quicum arcana, quicum occulta omnia?
3. An dolor longissimus quisque miserrimus, voluptatem non optabiliorem diuturnitas facit?
4. Multoque hoc melius nos veriusque quam Stoici.
5. Stuprata per vim Lucretia a regis filio testata civis se ipsa interemit.
6. Ego vero isti, inquam, permitto.

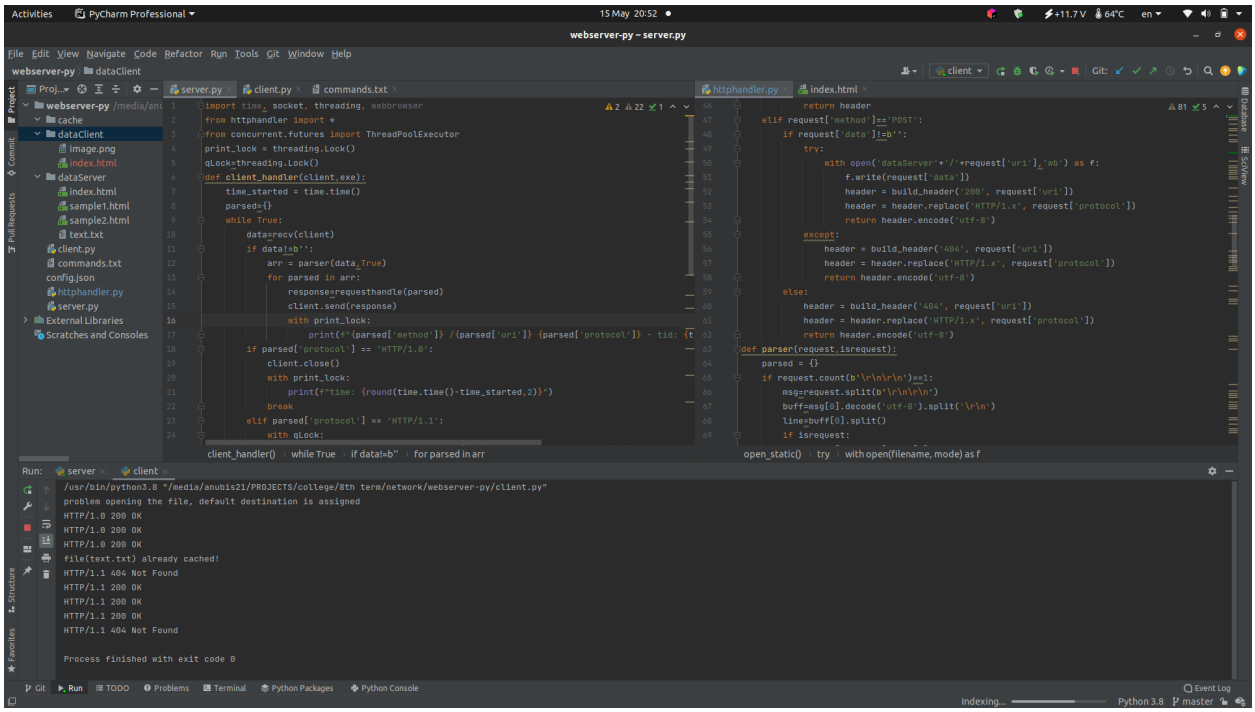
Graecum enim hunc versum nostis omnes-: Suavis laborum est praeteritorum memoria. Qui enim existimabit posse se miserum esse beatus non erit. Si qua in iis corrigere voluit, deteriora fecit. Si qua in iis corrigere voluit, deteriora fecit. Dic in quovis conventu te omnia facere, ne doleas. Tu quidem reddes;

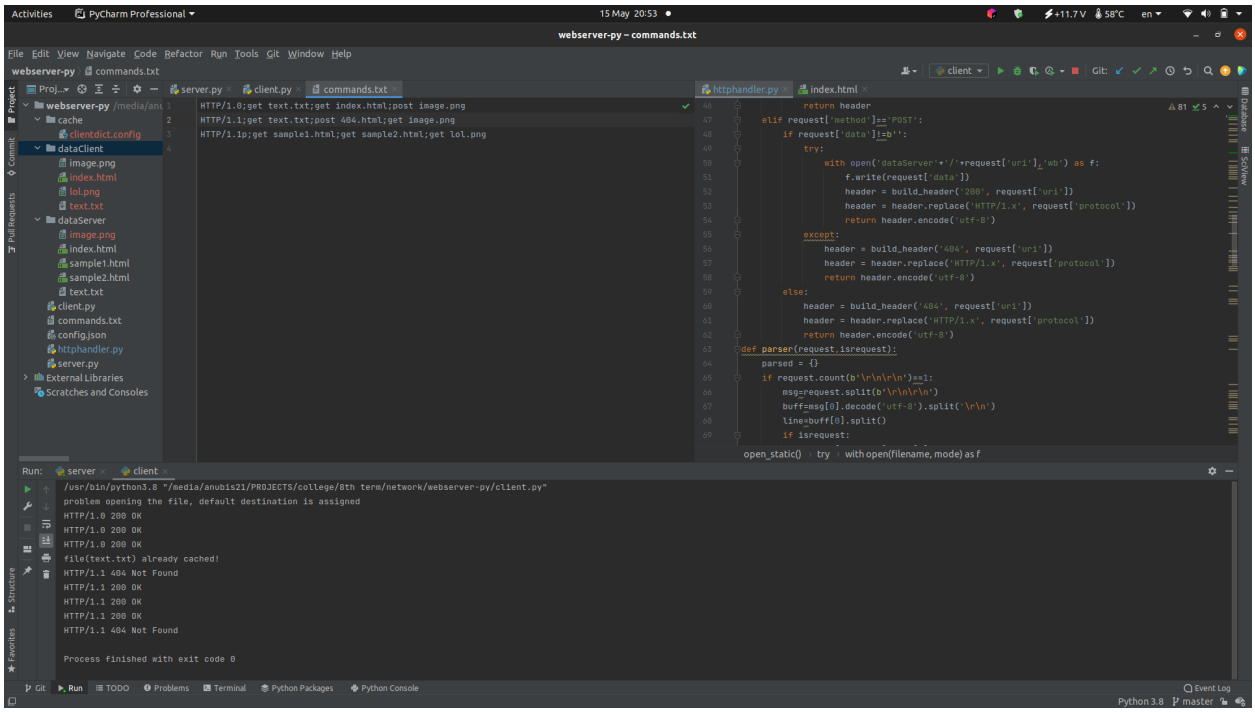
- Duo Reges: constructio interrete.
- Contineo me ab exemplis.
- Quo plebiscito decreta a senatu est consuli quaestio Cn.
- Quicquid porro animo cernimus, id omne oritur a sensibus;
- Eam si varietatem diceres, intellegerem, ut etiam non dicente te intellego;
- Qua ex cognitione facilior facta est investigatio rerum occultissimarum.

  Me igitur ipsum ames oportet, non mea, si veri amici futuri sumus.

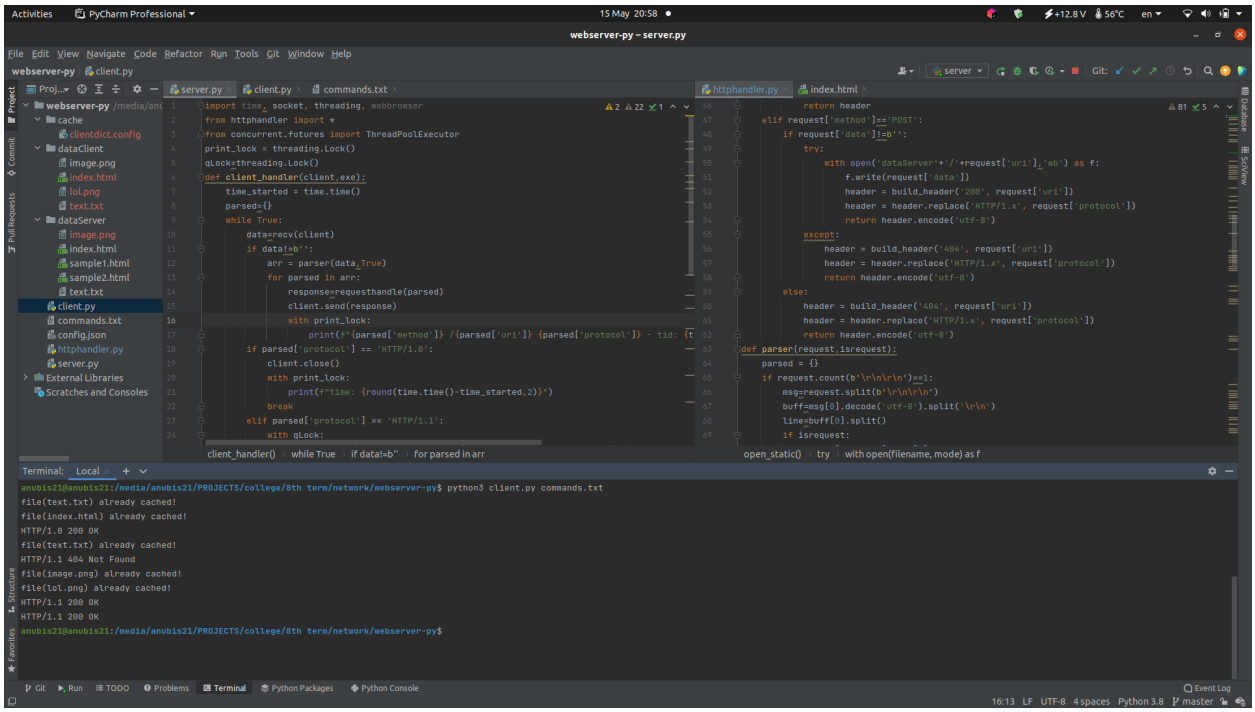
Testing client.py:







Testing client.py from the terminal with caching enabled:



FEATURES

- Thread pool
- Heuristics Adaptive to the number of sockets at the queue of the threading pool
- Cashing at client side
- Modular code
- Pipelining and persistent connections
- Parse multiple heads